B.Tech. First Year | Academic Year 2024–25

Department of Chemistry & Chemical Engineering

ENGINEERING CHEMISTRY

UNIT I – Complete Study Module

Topics Covered

Soft & Hard Water • Types of Hardness • EDTA Method • Dissolved Oxygen Estimation • Boiler Troubles • Scale & Sludge • Caustic Embrittlement • Drinking Water Standards • Ion-Exchange • Reverse Osmosis • Electrodialysis • Desalination

Subject Code	CHY 101 / ES-CH 101
Course	B.Tech – All Branches
Semester	I / II (First Year)
Unit	Unit I – Water Technology
Total Pages	25+ Pages (Exam-Oriented)
Prepared By	Department of Applied Chemistry

"This module is designed for B.Tech students as per the prescribed syllabus.

All diagrams are original SVG illustrations for exam preparation."

Table of Contents

Engineering Chemistry | Unit I – Water Technology

Syllabus Overview

Topic Cluster	Sub-Topics	Exam Weightage
Water Fundamentals	Hardness types, Soft/Hard water	High
Estimation Techniques	EDTA Method, DO Estimation	Very High
Boiler Troubles	Priming, Foaming, Scale, Sludge, Embrittlement	Very High
Water Treatment	Ion Exchange, RO, Electrodialysis	High
Standards	BIS & WHO Drinking Water Specs	Moderate

Exam Tip: Boiler troubles and EDTA method are asked almost every year. Diagrams for RO and Ion-Exchange carry direct marks. Learn all formulas from Section 4.

Section 1: Introduction to Water Technology

Engineering Chemistry | Unit I

1.1 Importance of Water in Industry

Water is one of the most abundant and universally used solvents on Earth. In engineering and industrial contexts, water serves as a coolant, solvent, steam generator, heat transfer medium, and raw material. However, natural water is never chemically pure—it contains dissolved salts, gases, organic matter, and suspended particles that make it unsuitable for direct industrial use.

The study of **Water Technology** involves understanding the properties of water, the impurities it contains, methods to detect and remove those impurities, and the treatment processes to make water fit for various uses including drinking, boiler feed, irrigation, and industrial processes.

1.2 Sources of Water

Source	Type	Main Impurities	Typical Use
Rainwater	Relatively pure	Dissolved CO_2, dust, SO_2	Collected for analysis
River Water	Surface water	Silt, clay, bacteria, organic matter	Public supply (after treatment)
Lake Water	Surface water	Algae, dissolved salts	Domestic, industrial
Well Water	Ground water	$CaCO_3$, $MgSO_4$, $FeSO_4$	Rural domestic
Sea Water	Saline	NaCl (3.5%), $MgCl_2$, SO_4 salts	After desalination

1.3 Types of Impurities in Water

- **Dissolved Impurities:** Salts of Ca, Mg, Na, Fe; dissolved gases like CO_2, O_2, H_2S
- **Suspended Impurities:** Sand, silt, clay, algae, bacteria
- **Colloidal Impurities:** Very fine particles of clay, silica, organic matter
- **Biological Impurities:** Bacteria, viruses, protozoa, fungi

1.4 Water Quality Parameters

The quality of water is assessed using several parameters:

- **Physical:** Colour, turbidity, taste, odour, temperature
- **Chemical:** pH, hardness, alkalinity, TDS (Total Dissolved Solids), DO (Dissolved Oxygen), BOD, COD

- **Biological:** Bacterial count, coliform count

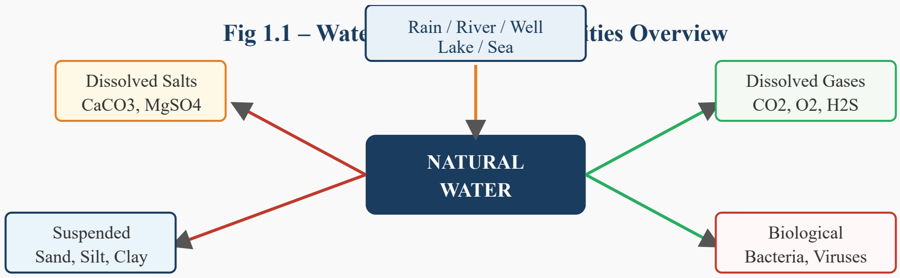

Fig. 1.1 – Types of Impurities in Natural Water Sources

Section 2: Soft Water and Hard Water

2.1 Definition

Soft Water: Water that lathers freely with soap and does not form a precipitate (scum) is called soft water. It contains very little or no dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) salts.

Hard Water: Water that does not produce lather easily with soap, but instead forms a white curdy precipitate (scum) is called hard water. The hardness is due to dissolved salts of Ca^{2+} and Mg^{2+} (bicarbonates, sulphates, chlorides and nitrates).

2.2 Cause of Hardness

When rainwater passes through soil and rocks, it dissolves various mineral salts. The primary salts responsible for hardness are:

- Calcium bicarbonate — $Ca(HCO_3)_2$
- Magnesium bicarbonate — $Mg(HCO_3)_2$
- Calcium sulphate — $CaSO_4$
- Magnesium sulphate — $MgSO_4$
- Calcium chloride — $CaCl_2$
- Magnesium chloride — $MgCl_2$
- Calcium nitrate — $Ca(NO_3)_2$
- Magnesium nitrate — $Mg(NO_3)_2$

2.3 Reaction of Soap with Hard Water

Soap is sodium stearate ($C_{17}H_{35}COONa$). When added to hard water, it reacts with Ca^{2+} and Mg^{2+} to form an insoluble white curd (scum) instead of lather.

Reaction with Calcium Salt:

```
2 C17H35COONa + CaCl2 → (C17H35COO)2Ca + 2 NaCl (sodium stearate)
(calcium stearate — insoluble scum)
```

Reaction with Magnesium Salt:

```
2 C17H35COONa + MgSO4 → (C17H35COO)2Mg + Na2SO4 (sodium stearate)
(magnesium stearate — insoluble)
```

2.4 Disadvantages of Hard Water

Area	Problem Caused
Domestic use	Does not lather easily; wastes soap; leaves scum on bathtubs
Boilers	Forms scale on inner walls; reduces heat transfer; causes explosions
Textile Industry	Affects dyeing; damages fabric fibres
Paper Industry	Affects paper quality; causes uneven finish
Food Industry	Changes taste of beverages; affects cooking quality
Leather Industry	Interferes with tanning process

Fig 2.1 – Hard Water vs Soft Water Comparison

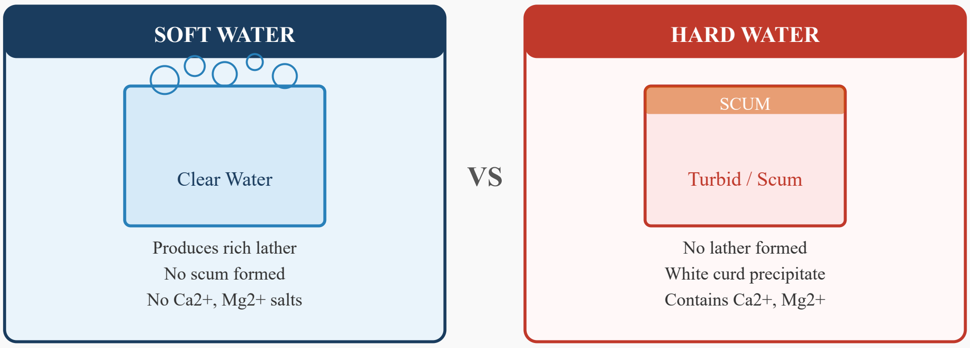

Fig. 2.1 – Visual Comparison: Hard Water vs Soft Water with Soap

Section 3: Types of Hardness

3.1 Temporary Hardness (Carbonate Hardness)

Definition: Hardness caused by dissolved bicarbonates of calcium [$Ca(HCO_3)_2$] and magnesium [$Mg(HCO_3)_2$] is called *Temporary Hardness*. It can be removed simply by boiling or by adding lime.

When water passes over limestone rocks, CO_2 dissolved in rainwater converts insoluble $CaCO_3$ into soluble $Ca(HCO_3)_2$:

Formation of Temporary Hardness:

```
CaCO3 + H2O + CO2 → Ca(HCO3)2 (insoluble) (soluble — causes hardness)
MgCO3 + H2O + CO2 → Mg(HCO3)2 (insoluble) (soluble — causes hardness)
```

Removal by Boiling:

```
Ca(HCO3)2 → (heat) CaCO3↓ + H2O + CO2↑ Mg(HCO3)2 → (heat) MgCO3↓ +
H2O + CO2↑ MgCO3 + H2O → Mg(OH)2↓ + CO2↑
```

3.2 Permanent Hardness (Non-Carbonate Hardness)

Definition: Hardness caused by dissolved sulphates, chlorides, and nitrates of calcium and magnesium is called *Permanent Hardness*. It CANNOT be removed by boiling; chemical treatment is required.

Salts responsible for permanent hardness:

- Calcium sulphate — $CaSO_4$
- Calcium chloride — $CaCl_2$
- Magnesium sulphate — $MgSO_4$
- Magnesium chloride — $MgCl_2$
- Calcium nitrate — $Ca(NO_3)_2$

3.3 Units of Hardness

Hardness is expressed in parts per million (ppm) or milligrams per litre (mg/L) as **$CaCO_3$ equivalent**.

Hardness Formula:

Hardness (ppm) = (Mass of hardness-causing salt / Volume of water in litres) x (Equivalent weight of CaCO3 / Equivalent weight of salt) x 1000 1 ppm = 1 mg/L = 1 g per 10^6 g of water 1 degree Clarke (Clark) = 1 grain per gallon (gpg) 1 degree French = 1 part CaCO3 per 10^5 parts water = 10 ppm

3.4 Comparison Table – Temporary vs Permanent Hardness

Property	Temporary Hardness	Permanent Hardness
Cause	Bicarbonates of Ca & Mg	Sulphates, Chlorides, Nitrates of Ca & Mg
Removal by Boiling	Yes – removed easily	No – cannot be removed
Also Called	Carbonate Hardness	Non-Carbonate Hardness
Salts Involved	$Ca(HCO_3)_2$, $Mg(HCO_3)_2$	$CaSO_4$, $CaCl_2$, $MgSO_4$, $MgCl_2$
Treatment Method	Boiling, Clark's Process (lime)	Soda-lime process, Ion-exchange
Scale Formation	Less severe	More severe (hard scale)
Sludge	Forms loose sludge	Forms hard adherent scale

Fig 3.1 – Classification of Hardness of Water

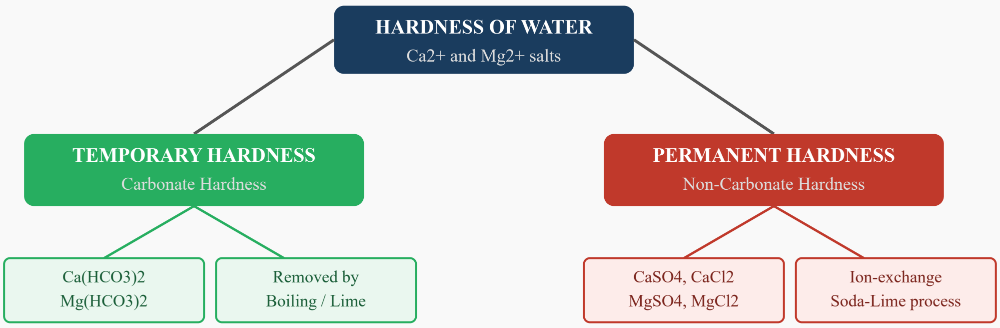

Fig. 3.1 – Classification Tree of Hardness of Water

Section 4: EDTA Method for Hardness Estimation

4.1 Principle

EDTA (Ethylene Diamine Tetraacetic Acid) is a complexometric titrant that forms stable, soluble 1:1 complexes with Ca^{2+} and Mg^{2+} ions. The disodium salt of EDTA (Na_2H_2Y) is commonly used for titrations.

At **pH 10** (maintained by NH_3/NH_4Cl buffer), EDTA reacts quantitatively with both Ca^{2+} and Mg^{2+} ions. The indicator used is **Eriochrome Black T (EBT)**.

4.2 Indicator (EBT) Behaviour

- Before endpoint: EBT forms a wine-red complex with Ca^{2+}/Mg^{2+} ions → Solution is **wine-red** colour.
- At endpoint: All Ca^{2+} and Mg^{2+} are taken up by EDTA; free EBT is released → Solution turns **blue**.

Reactions during EDTA Titration:

```
Step 1: Ca2+ + EBT (indicator) → [Ca-EBT] (Wine-red complex) Step 2: [Ca-EBT] + EDTA → [Ca-EDTA] + EBT (free — Blue colour) Step 3: Mg2+ + EDTA → [Mg-EDTA] Endpoint: Wine-red → Blue (permanent colour change)
```

4.3 Procedure – Step by Step

1. Pipette out **100 mL** of the water sample into a clean conical flask.
2. Add **5 mL** of NH_3/NH_4Cl buffer (pH = 10) to the flask.
3. Add **4-5 drops** of Eriochrome Black T (EBT) indicator. The solution turns wine-red.
4. Fill the burette with **standard EDTA solution** (0.01 M).
5. Add EDTA slowly from the burette while swirling the flask constantly.
6. Continue until the **endpoint** is reached — the colour changes from wine-red to **steel blue**.
7. Note the volume of EDTA consumed (V mL).
8. Repeat the titration two more times for concordant readings.

4.4 Calculation Formulas

Total Hardness Formula:

Total Hardness (ppm) = (V1 x M x 1000 x Molar mass of CaCO3) / Volume of sample (mL) Where: V1 = Volume of EDTA consumed for total hardness (mL) M = Molarity of EDTA solution (mol/L) Molar mass of CaCO3 = 100 g/mol Volume of sample = 100 mL (standard) Simplified: Total Hardness (ppm) = V1 x M x 1000 (when 100 mL sample, 0.01 M EDTA)

Permanent and Temporary Hardness:

Permanent Hardness (ppm) = (V2 x M x 1000 x 100) / Volume of boiled sample Temporary Hardness (ppm) = Total Hardness - Permanent Hardness Where V2 = volume of EDTA used for the boiled (filtered) sample

4.5 Numerical Example

Solved Problem:

Problem: 100 mL of a water sample required 20.0 mL of 0.01 M EDTA solution for total hardness titration. 100 mL of the boiled and filtered sample required 12.0 mL of the same EDTA. Calculate total, permanent, and temporary hardness.

Solution:

Given: Volume of EDTA for total hardness, V1 = 20.0 mL Volume of EDTA for permanent hardness, V2 = 12.0 mL Molarity of EDTA = 0.01 M Volume of sample = 100 mL Total Hardness = (20.0 x 0.01 x 1000 x 100) / 100 = 200 ppm (as CaCO3) Permanent Hardness = (12.0 x 0.01 x 1000 x 100) / 100 = 120 ppm Temporary Hardness = Total - Permanent = 200 - 120 = 80 ppm ANSWER: Total Hardness = 200 ppm, Permanent = 120 ppm, Temporary = 80 ppm

Fig 4.1 – EDTA Titration Setup

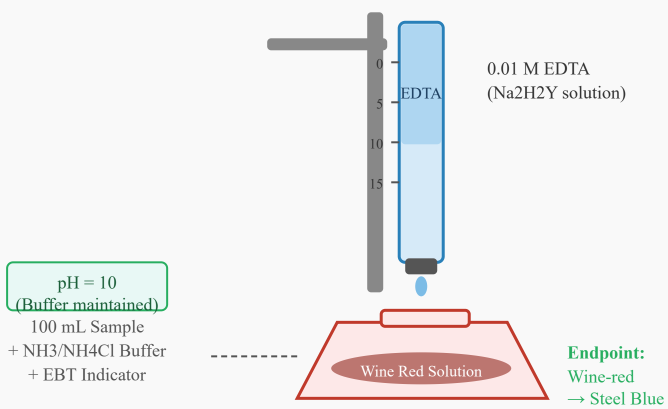

Fig. 4.1 – EDTA Complexometric Titration Setup for Hardness Estimation

Section 5: Estimation of Dissolved Oxygen (DO)

5.1 Significance of Dissolved Oxygen

Dissolved Oxygen (DO) is the amount of oxygen dissolved in water. It is a critical indicator of water quality:

- DO > 6 ppm: Water is suitable for aquatic life (fish, bacteria, etc.)
- DO < 4 ppm: Water quality is poor; aquatic life is stressed
- DO = 0 ppm: Anaerobic (septic) conditions; water is unfit
- In boiler water, DO causes corrosion of boiler tubes and pipes

5.2 Winkler's Method (Iodometric Method)

Principle: DO oxidises $Mn(OH)_2$ to $MnO(OH)_2$ in alkaline conditions. In acidic conditions, $MnO(OH)_2$ oxidises I^- to I_2. The liberated iodine is then titrated against standard sodium thiosulphate solution. The amount of iodine is equivalent to the DO.

5.3 Reagents Required

- Manganous sulphate ($MnSO_4$) solution
- Alkaline KI solution (KOH + KI)
- Concentrated H_2SO_4
- Standard sodium thiosulphate ($Na_2S_2O_3$) solution (0.025 N)
- Starch indicator

5.4 Reactions

Step 1 – Fixation of DO (Alkaline conditions):

```
MnSO4 + 2 KOH → Mn(OH)2 + K2SO4 (white ppt) 2 Mn(OH)2 + O2 → 2 MnO(OH)2 + H2O (brown ppt — DO fixed)
```

Step 2 – Release of I2 (Acidic conditions — add H2SO4):

```
MnO(OH)2 + 2 KI + H2SO4 → MnSO4 + I2 + 2 H2O + K2SO4 (I2 is liberated in proportion to DO)
```

Step 3 – Titration with Na2S2O3:

I2 + 2 Na2S2O3 → Na2S4O6 + 2 NaI (blue starch complex decolourises at endpoint) DO (mg/L) = (V x N x 8 x 1000) / Volume of sample in mL Where: V = Volume of Na2S2O3 used (mL) N = Normality of Na2S2O3 (0.025 N) 8 = Equivalent weight of oxygen

5.5 Numerical Example

Solved Problem:

Problem: In Winkler's titration, 200 mL of water sample required 5.0 mL of 0.025 N $Na_2S_2O_3$. Calculate the dissolved oxygen content.

Solution:

V = 5.0 mL N = 0.025 N Volume of sample = 200 mL DO = (5.0 x 0.025 x 8 x 1000) / 200 = (1000) / 200 = 5.0 mg/L ANSWER: DO = 5.0 mg/L (ppm) — Acceptable level for aquatic life

Fig 5.1 – Winkler's Method: Steps for Dissolved Oxygen Estimation

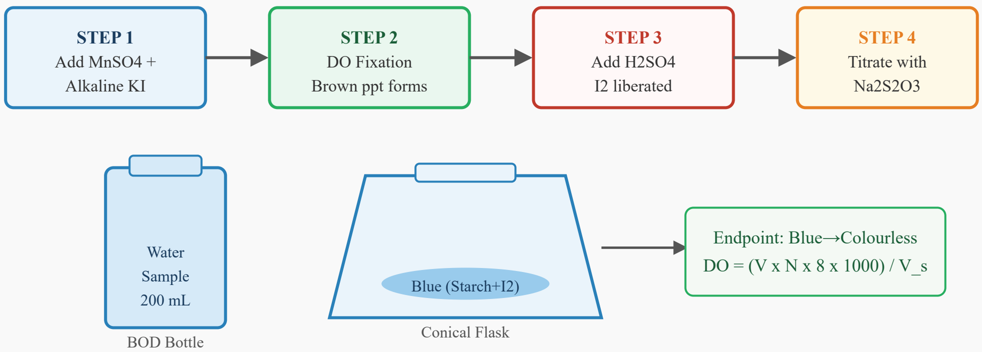

Fig. 5.1 – Step-by-Step Winkler's Method for Dissolved Oxygen Estimation

Section 6: Boiler Troubles – Priming and Foaming

6.1 Introduction to Boiler Feed Water

Boilers use water to generate high-pressure steam for power generation, industrial heating, and other applications. The water used must be purified, otherwise it causes serious operational troubles. The major boiler troubles are: **Priming, Foaming, Scale formation, Sludge, and Caustic Embrittlement**.

6.2 Priming

Definition: The process of carrying over of water droplets along with the steam from the boiler is called *Priming*. It is also known as "wet steam" problem.

Causes of Priming:

- Presence of dissolved salts ($NaCl$, Na_2SO_4) in large amounts
- Sudden increase in steam demand
- High water level in the boiler
- Improper boiler design
- Rapid heating of boiler water

Disadvantages of Priming:

- Wet steam reduces efficiency of turbines
- Dissolved salts in wet steam deposit on turbine blades, reducing efficiency
- Corrosion of turbine blades and superheater tubes
- Water hammer effect in pipes

Prevention: Maintain low water level, use water free of dissolved salts, slow heating, and use steam separators.

6.3 Foaming

Definition: Formation of persistent bubbles or foam on the surface of boiler water, which do not break easily, is called *Foaming*.

Causes of Foaming:

- Presence of oils, fats, and organic matter in boiler water
- Presence of alkalis and detergents

- Dissolved sodium compounds like Na_2CO_3 and NaOH

Disadvantages: Foam carries over with steam (similar to priming), deposits impurities on turbine blades, reduces steam quality.

Prevention: Remove oils/greases before boiler feed; use anti-foaming agents; add castor oil as an anti-foam agent; avoid excessive alkali treatment.

6.4 Comparison – Priming vs Foaming

Parameter	Priming	Foaming
Nature	Water droplets carried with steam	Foam/bubbles on water surface
Cause	High dissolved solids, rapid boiling	Oils, organic matter, alkalis
Result	Wet steam; turbine damage	Steam carryover; turbine fouling
Prevention	Steam separator; low water level	Remove oils; anti-foam agents

Fig 6.1 – Boiler Diagram Showing Priming and Foaming

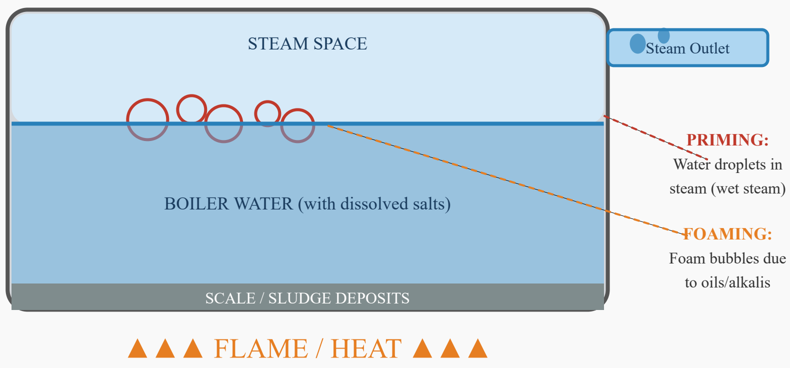

Fig. 6.1 – Cross-section of Boiler Showing Priming, Foaming, and Scale Deposits

Section 7: Scale and Sludge Formation

7.1 Scale Formation

Definition: The hard, adherent deposits formed on the inner walls of boilers due to the precipitation of dissolved salts (mainly $CaSO_4$, $CaCO_3$, $Mg(OH)_2$) during continuous evaporation of water are called *Scale*.

7.2 Causes of Scale Formation

As water evaporates in the boiler, the concentration of dissolved salts increases. When the solubility limit is exceeded, these salts precipitate on the hot inner surfaces:

Decomposition Reactions causing Scale:

```
Ca(HCO3)2 → (heat) CaCO3↓ + H2O + CO2↑ [Scale] Mg(HCO3)2 → (heat) MgCO3 + H2O + CO2 MgCO3 + H2O → Mg(OH)2↓ + CO2↑ [Scale] CaSO4 → (high temp) precipitates as hard scale (inverse solubility)
```

7.3 Types of Scale

Type	Composition	Nature	Formed at
Soft Scale	CaCO3, MgCO3, Mg(OH)2	Soft, easily removed	Low pressure boilers
Hard Scale	CaSO4 (inverse solubility)	Hard, glassy, firmly adherent	High temp. boilers
Silica Scale	SiO2, CaSiO3, MgSiO3	Very hard, difficult to remove	High pressure boilers

7.4 Disadvantages of Scale

- **Poor Heat Transfer:** Scale acts as an insulator (thermal conductivity of scale 30x less than iron). More fuel is consumed.
- **Overheating of Boiler Plates:** Metal softens due to excess heat retention, leading to buckling and blistering.
- **Boiler Explosions:** Thermal stress in scale-covered areas can cause explosions.
- **Reduced Efficiency:** Steam pressure drops; fuel costs increase by up to 30%.
- **Choking of Water Tubes:** Reduces water circulation.

7.5 Removal of Scale

- **Mechanical method:** Wire brushes, scrapers, pneumatic hammers
- **Thermal method:** Sudden cooling causes cracking of scale
- **Chemical method:** Treat with 5–10% HCl solution (for $CaCO_3$ scale); use EDTA for $CaSO_4$ scale

Chemical Removal of CaCO3 Scale:

```
CaCO3 + 2 HCl → CaCl2 + H2O + CO2↑
```

7.6 Sludge Formation

Definition: The soft, loose, slimy precipitate formed at the bottom of a boiler during evaporation of water is called *Sludge*. Unlike scale, sludge is non-adherent and can be removed by "blow-down" operation.

Composition of Sludge: $MgCO_3$, $MgCl_2$, $MgSO_4$, $CaCl_2$, and other soluble salts in high concentration.

Disadvantages of Sludge:

- If carried over with steam, it deposits on turbine blades
- Interferes with boiler efficiency if accumulated in large amounts
- Promotes corrosion at local concentration points

Removal: By periodic "blow-down" — draining the boiler water from the bottom to remove accumulated sludge.

7.7 Difference – Scale vs Sludge

Property	Scale	Sludge
Nature	Hard, adherent deposits	Soft, loose, non-adherent
Location	On boiler walls and tubes	At the bottom of boiler
Composition	CaSO4, CaCO3, Mg(OH)2	MgCO3, MgCl2, CaCl2
Heat transfer	Very poor — major problem	Less of a concern
Removal	Scraping, chemical treatment	Blow-down operation
Danger	Can cause boiler explosion	Reduces efficiency

Fig 7.1 – Scale Formation Process in Boiler Tubes

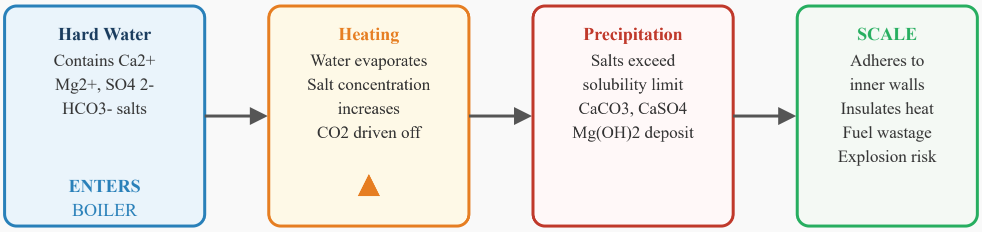

Fig. 7.1 – Scale Formation Process in Boiler Walls Step by Step

Section 8: Caustic Embrittlement

8.1 Definition

Caustic Embrittlement (also called *Stress Corrosion Cracking*) is the cracking of boiler steel at stressed areas such as rivets, joints, and bends due to the action of concentrated sodium hydroxide (NaOH). The boiler metal becomes brittle and develops hairline cracks, leading to failure.

8.2 Mechanism

In a boiler, sodium carbonate (Na_2CO_3) is sometimes added to soften feed water. At high temperature and pressure, Na_2CO_3 hydrolyses to form NaOH (caustic soda):

Formation of Caustic NaOH:

```
Na2CO3 + H2O → (high temp) 2 NaOH + CO2↑ The NaOH seeps into fine hairline cracks and stressed areas (rivets, bends, joints). NaOH dissolves the protective Fe3O4 layer on the iron surface: Fe3O4 + 4 NaOH → 2 NaFeO2 + Na2FeO2 + 4 H2O (protective layer destroyed; iron becomes reactive) Fe + 2 NaOH → Na2FeO2 + H2↑ (iron dissolves; H2 causes hydrogen embrittlement)
```

8.3 Conditions Favoring Caustic Embrittlement

- High concentration of NaOH (caustic soda) in boiler water
- Presence of stressed zones: rivets, bends, welds, and joints
- High temperature and pressure in the boiler
- Slow evaporation causing concentration of NaOH in crevices

8.4 Prevention of Caustic Embrittlement

- Add **sodium sulphate (Na_2SO_4)** or **tannin** to boiler water; these fill the pores and prevent NaOH seepage.
- Maintain the ratio $Na_2SO_4 : Na_2CO_3 = 1 : 1$ to $3 : 1$
- Avoid excess Na_2CO_3 treatment of feed water
- Use seamless welded boilers instead of riveted ones
- Keep boiler water alkalinity below 400 ppm (as NaOH)

8.5 Comparison – Boiler Troubles Summary

Trouble	Cause	Effect	Prevention
Priming	High TDS, rapid boiling	Wet steam, turbine damage	Steam trap, softened water
Foaming	Oils, alkalis, detergents	Foam carryover, deposits	De-oiling, anti-foam agents
Scale	$CaSO_4$, $CaCO_3$, $Mg(OH)_2$	Heat insulation, explosion	Softening, blow-down
Sludge	$MgCl_2$, $MgSO_4$, $CaCl_2$	Reduced efficiency	Blow-down operation
Caustic Embrittlement	Concentrated NaOH	Cracking of boiler metal	Na_2SO_4, tannin addition

Note: Caustic embrittlement is more common in riveted boilers than in welded ones. Modern boilers use seamless welded construction to minimise this problem.

Section 9: Industrial Water Treatment

9.1 Need for Water Treatment

Natural water contains impurities that make it unsuitable for industrial, domestic, and boiler use. Treatment processes are applied to remove or reduce these impurities to acceptable levels.

9.2 Water Treatment Flowchart for Public Supply

Fig 9.1 – Municipal Water Treatment Flowchart

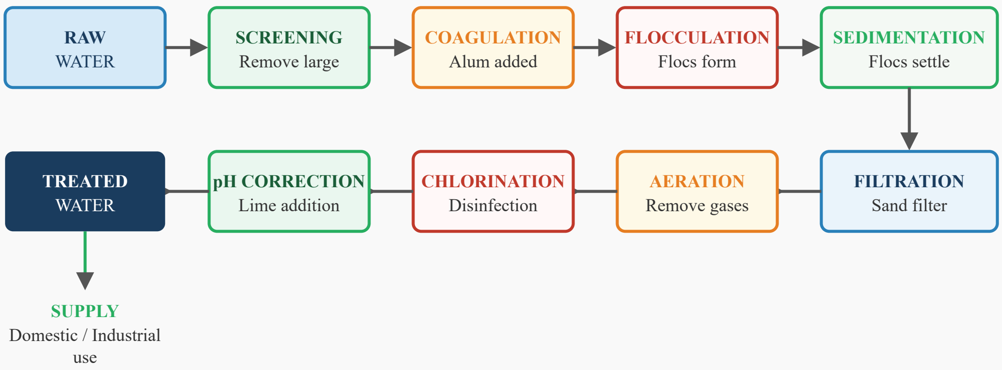

Fig. 9.1 – Complete Municipal Water Treatment Process Flowchart

9.3 Key Treatment Steps Explained

Step	Process	Chemical Used	Impurity Removed
Screening	Physical barrier (mesh/screen)	None	Large debris, leaves, fish
Coagulation	Addition of coagulant; neutralises charge of colloids	Alum [Al2(SO4)3]	Colloidal particles
Flocculation	Slow stirring; colloidal particles aggregate into flocs	Polyelectrolytes	Fine suspended matter
Sedimentation	Gravity settling of flocs in settling tank	None	Flocs, heavy particles
Filtration	Passage through sand/gravel bed	None	Fine particles, bacteria
Aeration	Spraying water into air or passing air through water	None	CO2, H2S, Fe2+, odour

Chlorination	Addition of Cl_2 or bleaching powder	Cl_2, $NaOCl$	Bacteria, viruses

Section 10: Drinking Water Standards – BIS and WHO

10.1 Need for Standards

Drinking water must be free from harmful contaminants. International and national standards define permissible limits for physical, chemical, and biological parameters to ensure public health safety.

10.2 BIS Standards (Bureau of Indian Standards – IS 10500:2012)

The BIS has specified two limits: **Desirable Limit** (preferred quality) and **Maximum Permissible Limit** (beyond which water is rejected).

Parameter	BIS Desirable Limit	BIS Max. Permissible	WHO Guideline
pH	6.5 – 8.5	No relaxation	6.5 – 8.5
Turbidity (NTU)	1	5	Less than 1 (ideal)
Total Hardness (ppm)	200	600	500
Total Dissolved Solids (ppm)	500	2000	1000
Chlorides (ppm)	250	1000	250
Sulphates (ppm)	200	400	250
Nitrates (ppm)	45	No relaxation	50
Fluorides (ppm)	1.0	1.5	1.5
Iron (ppm)	0.3	No relaxation	0.3
Manganese (ppm)	0.1	0.3	0.1 (health)
Copper (ppm)	0.05	1.5	2.0
Lead (ppm)	0.01	No relaxation	0.01
Arsenic (ppm)	0.01	No relaxation	0.01
Mercury (ppm)	0.001	No relaxation	0.006
Coliform count	Nil	Nil	0 per 100 mL
Dissolved Oxygen (ppm)	Desirable: 6+	—	Minimum 6

10.3 WHO Standards

The World Health Organization (WHO) publishes "Guidelines for Drinking Water Quality." These guidelines are adopted or adapted by most countries. WHO focuses on both chemical and microbiological safety:

- **Microbiological:** Total coliforms = 0 per 100 mL; E. coli = 0 per 100 mL
- **Radiological:** Total indicative dose < 0.1 mSv/year
- **Chemical:** Individual pesticide limits < 0.1 ug/L; total pesticides < 0.5 ug/L

Key Differences – BIS vs WHO:

BIS standards are legally enforceable in India. WHO guidelines are international recommendations. BIS provides "Desirable" and "Permissible" levels, while WHO provides single "Guideline Values." For most parameters, BIS and WHO values are comparable, but BIS chloride permissible limit (1000 ppm) is more relaxed than WHO (250 ppm).

Section 11: Ion-Exchange Process

11.1 Principle

Ion-Exchange Process is a water softening method in which the hard water is passed through ion-exchange resins (synthetic organic polymers) that exchange Ca^{2+} and Mg^{2+} ions with H^+ (from cation exchanger) and OH^- (from anion exchanger) ions, producing completely demineralised (deionised) water.

11.2 Types of Ion-Exchange Resins

Type	Symbol	Nature	Exchanged Ions
Cation Exchange Resin (RH)	RH	Acidic resin (–COOH, –SO3H groups)	Exchanges Ca2+, Mg2+, Na+ with H+
Anion Exchange Resin (ROH)	ROH	Basic resin (–NH2, =NH groups)	Exchanges Cl-, SO42-, HCO3- with OH-

11.3 Process – Step by Step

1. Hard water containing Ca^{2+}, Mg^{2+}, Cl^-, SO_4^{2-} is first passed through the **cation exchange column** (RH resin).
2. Ca^{2+} and Mg^{2+} ions are exchanged for H^+ ions. Water leaving is acidic and contains H^+, Cl^-, SO_4^{2-}.
3. Water then passes through the **anion exchange column** (ROH resin).
4. Cl^-, SO_4^{2-}, HCO_3^- ions are exchanged for OH^- ions.
5. H^+ and OH^- combine to form pure water (H_2O).
6. Result: Completely **demineralised** (deionised) water exits the system.

11.4 Chemical Reactions

Cation Exchange Column (RH):

2 RH + Ca2+ → R2Ca + 2 H+ 2 RH + Mg2+ → R2Mg + 2 H+ RH + Na+ → RNa + H+

Anion Exchange Column (ROH):

2 ROH + SO42- → R2SO4 + 2 OH- ROH + Cl- → RCl + OH- ROH + HCO3- → RHCO3 + OH-

Combination at outlet:

H+ + OH- → H2O (Pure water)

11.5 Regeneration of Resins

Cation Resin Regeneration (with dil. HCl or H2SO4):

R2Ca + 2 HCl → 2 RH + CaCl2 R2Mg + 2 HCl → 2 RH + MgCl2

Anion Resin Regeneration (with dil. NaOH):

R2SO4 + 2 NaOH → 2 ROH + Na2SO4 RCl + NaOH → ROH + NaCl

11.6 Advantages and Limitations

Advantages	Limitations
Produces completely demineralised water	High initial cost of resins
Removes all anions and cations	Colloidal matter clogs the resin
Resins are regenerable (reusable)	Turbid water must be filtered first
Compact design; easy to automate	Not suitable for brackish/sea water
pH of outlet is neutral	Organic matter can foul resins

Fig 11.1 – Ion-Exchange Demineralisation Setup

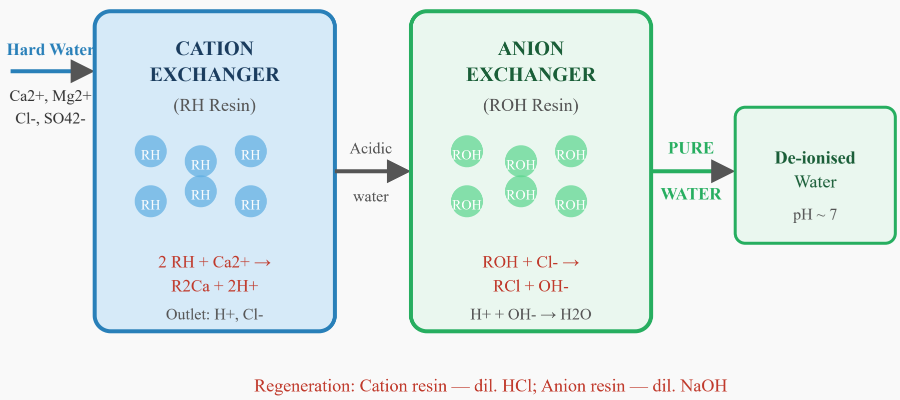

Fig. 11.1 – Ion-Exchange Demineralisation Plant with Cation and Anion Columns

Section 12: Desalination – Reverse Osmosis and Electrodialysis

12.1 Introduction to Desalination

Desalination is the process of removing dissolved salts (mainly NaCl) from sea water or brackish water to produce fresh drinking water. This is essential in water-scarce regions. Major methods are:

- Reverse Osmosis (RO)
- Electrodialysis (ED)
- Multi-Stage Flash Distillation (MSF)
- Multi-Effect Distillation (MED)

12.2 Reverse Osmosis (RO)

Osmosis: Natural movement of solvent (water) from a dilute solution to a concentrated solution through a semi-permeable membrane.

Reverse Osmosis: When a pressure greater than the osmotic pressure is applied to the concentrated (salt) side, water flows from the concentrated side to the dilute (fresh) side through the semi-permeable membrane. This is reverse osmosis.

Osmotic Pressure Formula (van't Hoff):

```
pi = MRT Where: pi = osmotic pressure (atm) M = molarity of the solution (mol/L) R = 0.0821 L·atm·mol^-1·K^-1 T = absolute temperature (K) For sea water: osmotic pressure ~ 25 atm Applied pressure in RO: 40 - 80 atm (must exceed osmotic pressure)
```

12.3 RO Membrane

- Material: Cellulose acetate or polyamide (thin-film composite)
- Pore size: 0.0001 micron (much smaller than bacteria, viruses, ions)
- Rejects: Salts, bacteria, viruses, heavy metals, nitrates, fluorides
- Passes: Water molecules

12.4 Process of Reverse Osmosis

1. Brackish/sea water is pre-treated (screening, filtration, pH adjustment) to remove suspended solids.
2. Pre-treated water is pressurised using a high-pressure pump (40–80 atm).
3. Pressurised water is passed through a semi-permeable RO membrane.
4. Water molecules pass through; dissolved salts are rejected and concentrated as "brine".
5. Fresh permeate water is collected and post-treated (pH correction, re-mineralisation).
6. Brine is discharged or further processed.

Fig 12.1 – Reverse Osmosis (RO) Desalination Setup

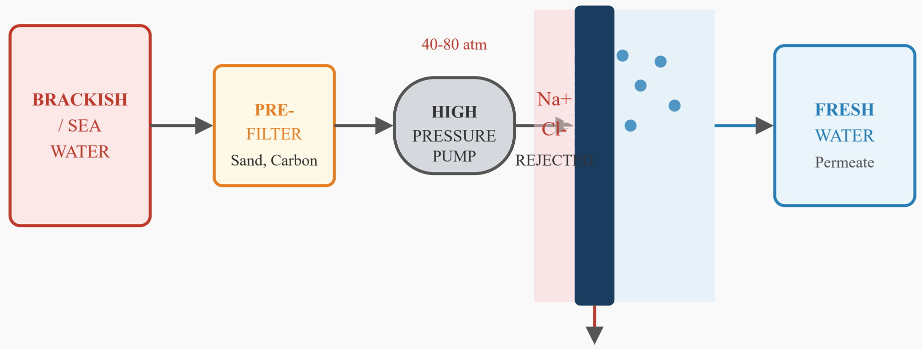

Fig. 12.1 – Reverse Osmosis (RO) Desalination System

12.5 Advantages and Disadvantages of RO

Advantages	Disadvantages
Removes 95–99% of dissolved salts	High energy consumption (high-pressure pump)
Removes bacteria, viruses, heavy metals	Membrane fouling — requires regular cleaning
No chemicals required	Pre-treatment of turbid water essential
Compact; suitable for small to large scale	Produces brine effluent (disposal problem)
Produces ultra-pure water	High membrane replacement cost

12.6 Electrodialysis (ED)

Electrodialysis is a membrane-based desalination process in which ions are transferred through ion-selective membranes under the influence of an applied electric potential (DC voltage). It is particularly suited for desalination of *brackish water* (1000–10,000 ppm TDS).

12.7 Working Principle

- An ED unit consists of alternating **cation-exchange membranes (CEM)** and **anion-exchange membranes (AEM)** between two electrodes (anode and cathode).
- CEM: Allows only cations (Na^{+}, Ca^{2+}, Mg^{2+}) to pass toward the cathode (negative).
- AEM: Allows only anions (Cl^{-}, SO_4^{2-}) to pass toward the anode (positive).
- Alternating chambers become dilute (fresh water) and concentrated (brine).
- Fresh water is collected from the dilute chambers.

12.8 Process

1. Brackish water is fed into all chambers between the electrodes.
2. DC voltage is applied across the electrodes.
3. Cations migrate toward the cathode, passing through CEM, but blocked by AEM in the next chamber.
4. Anions migrate toward the anode, passing through AEM, but blocked by CEM in the next chamber.
5. Alternating chambers are depleted of ions (dilute) or enriched (concentrate).
6. Fresh water is collected from dilute chambers; brine from concentrate chambers.

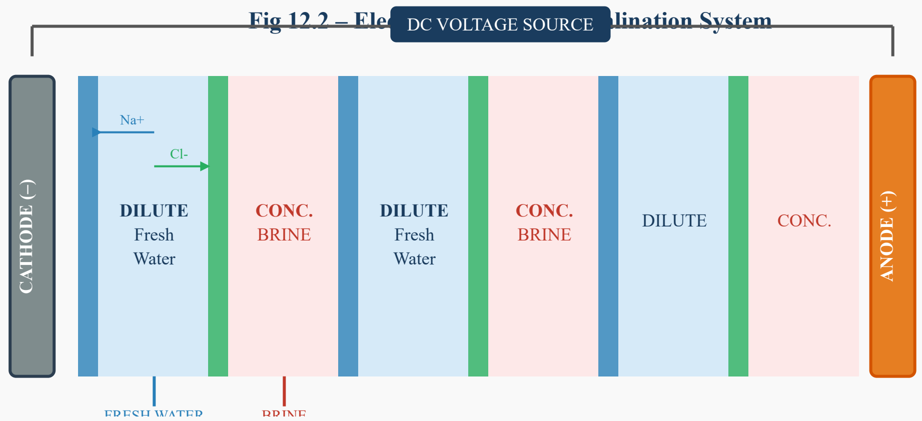

Fig. 12.2 – Electrodialysis (ED) Unit Showing Ion Migration Through Membranes

12.9 Comparison – RO vs Electrodialysis

Parameter	Reverse Osmosis (RO)	Electrodialysis (ED)
Driving Force	Applied hydraulic pressure (40–80 atm)	Applied electric potential (DC)
Mechanism	Pressure-driven membrane filtration	Electric-field-driven ion migration
Membrane Type	Semi-permeable (dense) membrane	Ion-selective membranes (CEM and AEM)
Suitable For	Sea water and brackish water	Brackish water (lower salinity)
Salt Removal	95–99% removal	50–90% removal (depending on stages)
Energy	High (high-pressure pumps)	Moderate (proportional to salt content)
Bacteria removal	Yes — physically excluded	No — only ions removed
Scale/Fouling	Membrane susceptible to fouling	Scale can foul membranes
Scale of Operation	Small to very large	Small to medium scale
Cost (capital)	Moderate to high	Moderate

Section 13: Questions and Answers

Part A – 2-Mark Questions (Short Answer)

Q1. Define hardness of water. 2 Marks

Hardness of water is the property by which water prevents the lathering action of soap and forms a white precipitate (scum) with soap. It is due to the presence of dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) salts in the form of bicarbonates, sulphates, chlorides, and nitrates. Hardness is measured in ppm (parts per million) as $CaCO_3$ equivalent. Hard water is unsuitable for domestic, industrial, and boiler use.

Q2. What is temporary hardness? How is it removed? 2 Marks

Temporary hardness is caused by dissolved bicarbonates of calcium [$Ca(HCO_3)_2$] and magnesium [$Mg(HCO_3)_2$]. It is also called carbonate hardness. It can be removed by simple boiling:

$Ca(HCO_3)_2 \rightarrow$ (heat) $CaCO_3 + H_2O + CO_2$

The $CaCO_3$ precipitates and can be filtered off. Lime ($Ca(OH)_2$) can also be used (Clark's process) to remove temporary hardness.

Q3. What is the principle of EDTA titration for hardness estimation? 2 Marks

EDTA (Ethylene Diamine Tetraacetic Acid) forms stable soluble 1:1 complexes with Ca^{2+} and Mg^{2+} ions at pH 10 (maintained by NH_3/NH_4Cl buffer). The indicator Eriochrome Black T (EBT) forms a wine-red complex with Ca^{2+}/Mg^{2+} initially. At the endpoint, all metal ions are sequestered by EDTA, EBT is released, and the solution turns steel blue. The volume of EDTA consumed is used to calculate hardness.

Q4. What is caustic embrittlement? 2 Marks

Caustic embrittlement (stress corrosion cracking) is the cracking of boiler metal at stressed zones (rivets, joints, bends) due to the action of concentrated sodium hydroxide (NaOH). NaOH is formed by hydrolysis of Na_2CO_3 at high temperature. It dissolves the protective Fe_3O_4 layer, causes dissolution of iron, and leads to hydrogen embrittlement, making the boiler metal brittle and prone to hairline cracking. Prevented by adding Na_2SO_4 or tannin to boiler water.

Q5. Differentiate between scale and sludge. 2 Marks

Scale is a hard, adherent deposit formed on the inner walls and heating tubes of a boiler due to the precipitation of $CaSO_4$, $CaCO_3$, and $Mg(OH)_2$. It acts as an insulator, reduces heat transfer, and can cause boiler explosions. Sludge is a soft, loose, non-adherent deposit formed at the bottom of the boiler, mainly composed of $MgCl_2$, $MgSO_4$, and $CaCl_2$. Sludge is less dangerous and is removed by blow-down operation. Scale is the more serious problem.

Q6. What is dissolved oxygen (DO)? What is its significance? 2 Marks

Dissolved oxygen (DO) is the amount of oxygen gas dissolved in water, measured in mg/L or ppm. It is a key indicator of water quality. High DO (above 6 ppm) indicates good water quality suitable for aquatic life. Low DO (below 4 ppm) indicates pollution. In boiler water, DO causes pitting corrosion of boiler tubes and pipes. Dissolved oxygen is measured by Winkler's (iodometric) method. It is removed from boiler feed water by mechanical deaeration or chemical treatment with sodium sulphite or hydrazine.

Part B – 5-Mark Questions (Medium Answer)

Q7. Explain the EDTA method for estimation of total hardness of water with reactions and calculations. 5 Marks

Principle: EDTA forms stable 1:1 complexes with Ca^{2+} and Mg^{2+} at pH 10. EBT indicator turns wine-red with these ions and steel blue at the endpoint when all ions are complexed by EDTA.

Reagents: (1) 0.01 M EDTA (standard solution), (2) NH_3/NH_4Cl buffer (pH 10), (3) EBT indicator.

Procedure: Pipette 100 mL of water sample into a conical flask. Add 5 mL of buffer and 4–5 drops of EBT indicator (solution turns wine-red). Titrate with 0.01 M EDTA from a burette until the colour changes from wine-red to steel blue.

Reactions:

At start: Ca^{2+} + EBT → [Ca-EBT] (wine-red)

During titration: [Ca-EBT] + EDTA → [Ca-EDTA] + EBT (free = blue)

Calculation:

Total Hardness (ppm) = (V_1 × M × 100 × 1000) / Volume of sample (mL)

If V_1 = 20 mL, M = 0.01 M, sample = 100 mL: Hardness = (20 × 0.01 × 100 × 1000) / 100 = 200 ppm

Temporary Hardness: Titrate boiled and filtered sample; V_2 = 12 mL. Permanent Hardness = (12 × 0.01 × 100 × 1000)/100 = 120 ppm. Temporary Hardness = 200 – 120 = 80 ppm.

Q8. Explain boiler troubles with emphasis on scale formation and its prevention. 5 Marks

Boilers are affected by several troubles related to impurities in feed water. The most damaging is **scale formation**.

Scale Formation: As water evaporates continuously in a boiler, the concentration of dissolved Ca^{2+} and Mg^{2+} salts increases. When the solubility limit is exceeded, these salts precipitate as hard deposits on the inner walls:

$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$

$Mg(HCO_3)_2 \rightarrow MgCO_3 + H_2O + CO_2 \rightarrow Mg(OH)_2 + CO_2$

$CaSO_4$ becomes less soluble at high temperature (inverse solubility) and forms hard scale.

Effects: Scale is a poor conductor of heat (thermal conductivity is 30 times lower than iron). This causes: (1) overheating of boiler plates, (2) warping and blistering of metal, (3) reduced efficiency and increased fuel cost, (4) in extreme cases, boiler explosions.

Prevention: (1) Soften water before boiler feed using ion exchange or lime-soda process. (2) Add scale inhibitors (e.g., sodium hexametaphosphate — calgon). (3) Perform regular blow-down operations. (4) Use internal chemical treatment (adding Na_2CO_3, NaOH, or tannin).

Removal: Mechanical scraping; thermal shocking; chemical treatment with dilute HCl for $CaCO_3$ scale.

Part C – 10-Mark Questions (Long Answer)

Q9. Explain in detail the ion-exchange process for water softening with a diagram, reactions, and regeneration. 10 Marks

Introduction: Ion exchange is one of the most effective methods for producing demineralised (deionised) water. It removes all dissolved cations (Ca^{2+}, Mg^{2+}, Na^{+}) and anions (Cl^{-}, SO_4^{2-}, HCO_3^{-}) from water by exchanging them with H^{+} and OH^{-} ions respectively.

Resins Used: (1) *Cation Exchange Resin (RH)*: A synthetic polymer with acidic groups (–COOH or $–SO_3H$) that can release H^{+} ions. (2) *Anion Exchange Resin (ROH)*: A synthetic polymer with basic groups ($–NH_2$ or =NH) that can release OH^{-} ions.

Cation Exchanger Reactions:

$2\,RH + Ca^{2+} \rightarrow R_2Ca + 2\,H^{+}$

$2\,RH + Mg^{2+} \rightarrow R_2Mg + 2\,H^{+}$

$RH + Na^{+} \rightarrow RNa + H^{+}$

Anion Exchanger Reactions:

$2\,ROH + SO_4^{2-} \rightarrow R_2SO_4 + 2\,OH^{-}$

$ROH + Cl^{-} \rightarrow RCl + OH^{-}$

$ROH + HCO_3^{-} \rightarrow RHCO_3 + OH^{-}$

Net Reaction: $H^{+} + OH^{-} \rightarrow H_2O$ (pure water — pH ~7)

Process Flow: Hard water → Pre-filter → Cation Exchanger → (acidic water) → Anion Exchanger → Deionised Water

Regeneration: After exhaustion of resins:

Cation resin: $R_2Ca + 2\,HCl \rightarrow 2\,RH + CaCl_2$ (pass dilute HCl)

Anion resin: $R_2SO_4 + 2\,NaOH \rightarrow 2\,ROH + Na_2SO_4$ (pass dilute NaOH)

Applications: Boiler feed water, pharmaceutical water, high-purity laboratory water, semiconductor manufacturing, power plant water treatment.

Advantages: Produces completely demineralised water; resins are regenerable; no chemicals in the product water; compact design.

Limitations: Turbid water must be pre-filtered; organic compounds can foul resins; high initial cost; not suitable for very saline waters.

Q10. Explain reverse osmosis and electrodialysis with diagrams and compare them.

10 Marks

Reverse Osmosis (RO): In normal osmosis, water flows from dilute to concentrated solution through a semi-permeable membrane until osmotic equilibrium is reached. In RO, a pressure greater than the osmotic pressure (40–80 atm for sea water) is applied to the concentrated side, forcing water molecules to flow against the osmotic gradient — from concentrated (saline) to dilute (fresh water) side. The membrane rejects salts, bacteria, viruses, and heavy metals. Only water molecules (size 0.0003 nm) pass through the membrane pores (~0.0001 micron).

Osmotic Pressure Formula: pi = MRT, where M = molarity, R = 0.0821 $L \cdot atm \cdot mol^{-1} \cdot K^{-1}$, T = absolute temperature. For sea water, pi ≈ 25 atm; applied pressure in RO = 40–80 atm.

RO Membranes: Cellulose acetate or polyamide thin-film composites. Arranged in spiral-wound modules for maximum surface area.

Electrodialysis (ED): ED uses ion-selective membranes arranged alternately (CEM and AEM) between two electrodes. When DC voltage is applied, cations migrate toward the cathode through CEMs, and anions migrate toward the anode through AEMs. Alternating chambers become depleted of ions (fresh water) or enriched (brine). ED is best suited for brackish water (1,000–10,000 ppm TDS).

Key Comparison Points: (1) RO uses pressure; ED uses electricity. (2) RO is better for sea water; ED is better for brackish water. (3) RO removes bacteria; ED does not remove bacteria (only ions). (4) Both produce brine effluent that must be disposed of. (5) ED energy cost is proportional to salt content; RO energy is less sensitive to salinity variation.

Applications of RO: Municipal water supply, desalination plants (Saudi Arabia, UAE), pharmaceutical, beverage industry, dialysis water.

Applications of ED: Desalination of brackish well water, production of table salt from sea water, concentration of whey in food processing, treatment of wastewater from electroplating.

Section 14: Quick Revision Sheet

One-page summary of all key formulas, reactions, and concepts for exam preparation.

Key Formulas

- Total Hardness (ppm) = (V_1 × M × 100 × 1000) / Sample vol. (mL)
- Temp. Hardness = Total – Permanent Hardness
- DO (mg/L) = (V × N × 8 × 1000) / Sample vol. (mL)
- Osmotic Pressure: pi = MRT
- 1 ppm = 1 mg/L; 1 French degree = 10 ppm
- EDTA: 0.01 M; Buffer: pH 10; Indicator: EBT

Important Reactions

- Temp. hardness: $Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$
- Perm. hardness: $CaSO_4$ → hard scale
- EDTA endpoint: wine-red → steel blue
- DO fixation: $Mn(OH)_2 + O_2 \rightarrow MnO(OH)_2$
- DO titration: $I_2 + 2\ Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2\ NaI$
- Caustic: $Fe + 2\ NaOH \rightarrow Na_2FeO_2 + H_2$

Boiler Troubles Summary

- **Priming:** Water droplets in steam; cause: high TDS, rapid boiling; prevent: steam separator
- **Foaming:** Foam on water surface; cause: oils/alkalis; prevent: anti-foam agents
- **Scale:** $CaSO_4$, $CaCO_3$; insulation; explosion risk
- **Sludge:** $MgCl_2$, $MgSO_4$; loose deposit; remove by blow-down
- **Embrittlement:** NaOH in crevices; cracking of metal; prevent: Na_2SO_4

Treatment Methods Summary

- **Ion Exchange:** Cation (RH) + Anion (ROH) columns; gives deionised water; regenerate with HCl and NaOH
- **RO:** Pressure (40–80 atm) through semi-permeable membrane; removes 95–99% salts
- **Electrodialysis:** DC voltage; CEM + AEM membranes; brackish water (1000–10000 ppm)
- **Chlorination:** Cl_2 or NaOCl; disinfection; 0.2 ppm residual Cl
- **Coagulation:** Alum [$Al_2(SO_4)_3$]; removes colloids

BIS Drinking Water Standards (Key)

- pH: 6.5 – 8.5
- Total Hardness: 200 ppm (max 600)
- TDS: 500 ppm (max 2000)
- Chlorides: 250 ppm (max 1000)
- Fluorides: 1.0 ppm (max 1.5)
- Nitrates: 45 ppm
- Arsenic: 0.01 ppm (no relaxation)
- Coliform: Nil

Types of Hardness – Quick Recap

- **Temporary:** $Ca(HCO_3)_2$, $Mg(HCO_3)_2$; removed by boiling
- **Permanent:** $CaSO_4$, $CaCl_2$, $MgSO_4$, $MgCl_2$; not removed by boiling
- **Total:** Temp. + Permanent hardness
- Units: ppm = mg/L (as $CaCO_3$)
- 1 French degree = 10 ppm

- EDTA factor = Mol. wt. of $CaCO_3$ = 100

Memory Tips (Mnemonics)

Concept	Mnemonic / Quick Tip
Boiler Troubles	**P F S S C** – Priming, Foaming, Scale, Sludge, Caustic embrittlement
EDTA Endpoint	"Wine goes Blue at the End" (Wine-red → Steel-Blue)
Temp. Hardness Salts	"BiCa BiMg" — Bicarbonates of Ca and Mg
Perm. Hardness Salts	"SCN for Ca and Mg" — Sulphates, Chlorides, Nitrates
RO vs ED	RO = Pressure-driven (pump); ED = Electricity-driven (voltage)
Winkler's DO Steps	"Fix Acidify Titrate" — Fix O2 with MnSO4, add H2SO4, titrate I2 with Na2S2O3
Ion Exchange	"RH eats cations; ROH eats anions" — simple way to remember resin function

Section 15: Textbook and Reference Books

Prescribed Textbooks

[1] **Jain P.C. and Jain M.** — *Engineering Chemistry*, 16th Edition, Dhanpat Rai Publishing Company, New Delhi (2013).

Primary reference for Unit I – Water Technology; covers hardness, boiler troubles, EDTA method, and water treatment comprehensively.

[2] **Atkins P., de Paula J., and Keeler J.** — *Atkins' Physical Chemistry*, 10th Edition, Oxford University Press, New York (2010).

Reference for thermodynamic principles including osmotic pressure, van't Hoff equation, and solution chemistry.

[3] **Shashi Chawla** — *A Textbook of Engineering Chemistry*, Dhanpat Rai & Co., New Delhi (2015).

Useful supplementary text for boiler chemistry and industrial water treatment.

[4] **O.G. Palanna** — *Engineering Chemistry*, Tata McGraw Hill, New Delhi (2009).

Good reference for desalination processes, RO, and electrodialysis.

Reference Books

[5] **H.F.W. Taylor** — *Cement Chemistry*, 2nd Edition, Thomas Telford Publishing, London (1997).

Reference for chemistry of calcium silicates and calcium-based deposits (scale).

[6] **D.J. Shaw** — *Introduction to Colloids and Surface Chemistry*, 4th Edition, Butterworth-Heinemann (1992).

Background reading for colloidal impurities in water, coagulation, and flocculation chemistry.

[7] **Fred W. Billmeyer Jr.** — *Textbook of Polymer Science*, 3rd Edition, John Wiley & Sons (1984).

Reference for synthetic polymer science applicable to ion-exchange resin structure and properties.

[8] **Bureau of Indian Standards (BIS)** — *Drinking Water Specification IS 10500:2012*, Bureau of Indian Standards, New Delhi (2012).

Official standard document for drinking water quality parameters and permissible limits in India.

[9] **World Health Organization (WHO)** — *Guidelines for Drinking-Water Quality*, 4th Edition with 1st Addendum, WHO Press, Geneva (2017).

International guideline values for chemical, microbiological, and radiological parameters in drinking water.

Online Resources

- NPTEL Video Lectures – Engineering Chemistry (IIT level): *nptel.ac.in*
- WHO Water Sanitation and Hygiene: *who.int/water_sanitation_health*

- BIS Standards Portal: *bis.gov.in*
- US EPA – Water Quality Criteria: *epa.gov/wqc*

END OF UNIT I – WATER TECHNOLOGY

Engineering Chemistry | B.Tech. 1st Year

Prepared with complete syllabus coverage, exam-oriented Q&A, and original SVG diagrams.

All formulas, reactions, and standards are as per the latest curriculum and BIS/WHO specifications.

Best of Luck for Your Examinations!